The total duty cycle devoted to balancing is approximately 70% per 250 ms. This is because a portion of the 250 ms is allotted for normal cell voltage measurements through the ADC.

If *[ADC_EN]* =1, OV and UV protections are not affected by cell balancing, since the cell balancing is temporarily suspended for a small slice of time every 250 ms during which the cell voltage readings are taken. This ensures that the OV and UV protections do not accidentally trigger, or miss an actual OV/UV condition on the cells while balancing is enabled.

备注

All cell balancing control bits in CELLBAL1, CELLBAL2, and CELLBAL3 are automatically cleared under the following events, and must be explicitly rewritten by the host microcontroller following clearing of the event:

- DEVICE_XREADY is set
- Enters NORMAL mode from SHIP mode

8.3.1.3.4 Alert

The ALERT pin serves as an active high digital interrupt signal that can be connected to a GPIO port of the host microcontroller. This signal is an OR of all bits in the SYS_STAT register.

In order to clear the ALERT signal, the source bit in the SYS_STAT register must first be cleared by writing a “1” to that bit. This will cause an automatic clear of the ALERT pin once all bits are cleared.

The ALERT pin may also be driven by an external source; for example, the pack may include a secondary overvoltage protector IC. When the ALERT pin is forced high externally while low, the device will recognize this as an OVRD_ALERT fault and set the *[OVRD_ALERT]* bit. This triggers automatic disabling of both CHG and DSG FET drivers. The device cannot recognize the ALERT signal input high when it is already forcing the ALERT signal high from another condition.

The ALERT pin has no internal debounce support so care should be taken to protect the pin from noise or other parasitic transients.

备注

It is highly recommended to place an external 500 kΩ – 1 MΩ pull-down resistor from ALERT to VSS as close to the IC as possible. Additional recommendations are:

a) To keep all traces between the IC and components connected to the ALERT pin very short.

b) To include a guard ring around the components connected to the ALERT pin and the pin itself.

8.3.1.3.5 Output LDO

An adjustable output voltage regulator LDO is provided as a simple way to provide power to additional components in the battery pack, such as the host microcontroller or LEDs. The LDO is configured in EEPROM by TI during the production test process, and can support 2.5-V or 3.3-V options.

A cascode small-signal FET must be added in the external path between BAT and REGSRC with the BQ76930 and BQ76940. This helps drop most of the power dissipation outside of the package and cuts down on package power dissipation.

8.3.1.4 Communications Subsystem

The AFE implements a standard 100-kHz I^2C interface and acts as a slave device. The I^2C device address is 7-bits and is factory programmed. Consult the Device Comparison Table (节 5) of this data sheet for more information.

A write transaction is shown in 图 8-3. Block writes are allowed by sending additional data bytes before the Stop. The I^2C block will auto-increment the register address after each data byte.